ugp-lean: A Machine-Checked Formalization of the Universal Generative Principle

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Abstract

We present **ugp-lean**, a LEAN 4 formalization of the Universal Generative Principle (UGP), a deterministic arithmetic framework defined over integer ridges $R_n = 2^n - 16$. The formalization covers: the ridge sieve, prime-lock criterion, Residual Seed Uniqueness (RSUC), Quarter-Lock identity, Generative Triple Evolution (GTE) canonical orbit, UWCA Turing universality, and a new class of primes (*UGP primes*) with ten machine-checked instances. The artifact comprises 48 LEAN 4 modules, builds against Mathlib v4.29.0-rc6, and maintains a strict *zero-sorry, zero-custom-axiom* policy on the core UGP proof path. Novel contributions include: a machine-checked proof that the GTE update map T *forces* the canonical orbit, a general ridge remainder lock for all $n \geq 5$, an even-step c-invariance theorem, a formal definition of UGP primes with existence proof and ten verified instances across three ridge levels, a precise statement of the mirror-dual conjecture with a proof that the divisor count $\tau(2^n - 16)$ is unbounded, an exact divisor-count formula $\tau(2^n - 16) = 5 \cdot \tau(2^{n-4} - 1)$ for all $n \geq 5$, and a new *mirror shift algebra*: a machine-checked proof that for any mirror-dual pair (b_2, q_2), both c_1-values are congruent to $t = 20$ modulo $b_1 = b_2 + q_2 + 7$, and their difference equals $b_1 \cdot (q_2 - b_2)$—a universal algebraic law holding at every ridge level. The repository also contains companion modules (`InertPrimes`, `AnalyticArchitecture`) providing citable analytic infrastructure derived from the UGP factory constants; these are documented in `docs/RESONANT_FACTORY_NOTE.md` and not discussed in this paper. All definitions are self-contained; all open conjectures are formally stated as LEAN 4 propositions.

Trust boundary: Proved statements are scoped to the ridged arithmetic framework formalized here; open conjectures remain explicit LEAN 4 `Prop`s and are not asserted as classical theorems.

1 Premises, disagreement coordinates, and trust boundary

Where disagreement can (and cannot) live

Formal spine. Classification and forcing lemmas refer to definitions in `Ugcp/` (or as cited in the module tree); numerical witnesses are `#eval`-friendly where stated.

Premise coordinates (for external readers).

- **Ridge arithmetic vs. ambient set theory.** Claims are about the constructed UGP objects (ridges, sieve, T-orbit), not about every conceivable number-theoretic universe. *Logical effect:* Generalizing beyond the formalized ridge family requires new definitions, not silent extension.
- **Toolchain / Mathlib pin.** The paper reports builds against a stated Mathlib revision; upgrading may require port work. *Logical effect:* Port failures do not erase prior checked revisions.

Keywords: interactive theorem proving, Lean 4, formalization, number theory, ridge sieve, Mersenne numbers, prime-producing formulas, cellular automata

2 Introduction

The Universal Generative Principle (UGP) is a deterministic number-theoretic framework that generates structured integer triples from a single arithmetic seed. The framework centers on

ridges $R_n = 2^n - 16$, a divisor-based sieve that selects *survivor* pairs, and a parity-dependent update map T called the Generative Triple Evolution (GTE) that evolves triples through an orbit. At the distinguished level $n = 10$, the sieve produces exactly two survivor pairs, and the Minimum Description Length (MDL) principle selects a unique seed—the *canonical seed* $(1, 73, 823)$—whose orbit under T is $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$.

The UGP framework makes strong arithmetic claims: uniqueness of the residual seed, primality of specific values, determinism of the orbit, and Turing universality of the substrate. These claims benefit from machine-checking for three reasons:

1. The number-theoretic arguments involve explicit computation (sieving divisors, verifying primality) that is error-prone by hand.
2. The framework introduces novel definitions (ridge sieve, Quarter-Lock, mirror-dual pairs) whose internal consistency is non-trivial.
3. A machine-checked artifact serves as a citable, independently verifiable record.

Contributions. The **ugp-lean** artifact formalizes the UGP framework in LEAN 4 [4] with Mathlib [7]. All definitions are self-contained: the formalization defines every concept it uses (ridge, sieve, update map, orbit, UGP primes) from first principles, depending only on Mathlib's standard number theory. Beyond the core framework, the formalization contributes several *new* results:

- **Orbit from T** (§6): The canonical orbit values are *derived* from the update map, not postulated.
- **General ridge remainder lock** (§7): For *all* $n \geq 5$ and all valid divisors b_2, $(2^n - 1) \bmod b_2 = 15$.
- **Even-step c-invariance** (§6): Under the strict per-step rule, the even step immediately after a ridge hit leaves c unchanged.
- **ugp primes** (§8): A new class of primes with a formal predicate, an existence proof, and ten machine-checked instances across three ridge levels.
- **Mirror-dual conjecture** (§11): A precise formal statement with five machine-verified witness pairs, a proof that the divisor count $\tau(2^n - 16)$ is unbounded, and an exact formula $\tau(2^n - 16) = 5 \cdot \tau(2^{n-4} - 1)$ via coprimality and multiplicativity of τ.
- **Resonant factory** (§11.1): Machine-checked algebraic infrastructure for a twin-prime program: branch linearization, gap-2 factory quadratics, and a Hasse check (no local obstruction at primes ≤ 43).
- **Mirror shift algebra** (§12): A new universal algebraic law: for every mirror-dual pair (b_2, q_2) at any ridge level, both c_1-values satisfy $c_1 \equiv 20 \pmod{b_1}$, and the shift $c_1(b_2, q_2) - c_1(q_2, b_2) = b_1 \cdot (q_2 - b_2)$. At $n = 10$ this gives $2137 - 823 = 73 \cdot 18 = 1314 = 2 \cdot 3^2 \cdot 73$. Machine-checked at $n = 10$, 13, 16.
- **Real analysis** (§13): The c_1 function on the divisor hyperbola is differentiable with derivative $13R/q^2 + 2q - 6 \geq 22$ on the valid shell—a machine-checked uniform magnification bound for sieve applications.

Structure. §3 defines the UGP framework. §4 describes the module architecture and design decisions. §5 presents the core classification results (RSUC). §6 covers the GTE orbit and update map. §7 collects the general-level theorems. §8 introduces UGP primes. §9 treats the Quarter-Lock and Elegant Kernel. §10 addresses universality and self-reference. §11 develops the mirror-dual conjecture. §12 presents the mirror shift algebra. §13 presents the real-analysis layer. §14 resolves seven conjectures and states four open ones. §15 surveys related work. §16 concludes.

3 The UGP Framework

We define the mathematical content that is formalized. All definitions below have direct Lean 4 counterparts in the `UgpLean.Core` modules.

Definition 3.1 (Ridge)**.** The *ridge* at level n is $R_n := 2^n - 16$. At $n = 10$, $R_{10} = 1008$.

Definition 3.2 (Strict-ridge divisor pair)**.** A *strict-ridge divisor pair* at level n is a pair $(b_2, q_2) \in \mathbb{N}^2$ with $b_2 \cdot q_2 = R_n$ and $b_2 \geq 16$, $q_2 \geq 16$.

From a divisor pair (b_2, q_2) the UGP derives:

$$b_1 = b_2 + q_2 + 7, \tag{1}$$
$$q_1 = q_2 - 13, \tag{2}$$
$$c_1 = b_1 \cdot q_1 + 20. \tag{3}$$

The constants $s = 7$, $g = 13$, $t = 20$ are the UGP-1 parameters.

Definition 3.3 (Prime-lock criterion)**.** A divisor pair (b_2, q_2) is *prime-locked* if c_1 as defined in (3) is prime.

At $n = 10$, $R_{10} = 1008$ has eight divisor pairs with both components ≥ 16. Of these, exactly two are prime-locked: $(24, 42)$ yielding $c_1 = 823$ (prime), and $(42, 24)$ yielding $c_1 = 2137$ (prime). The six non-survivors have composite c_1 values, each explicitly excluded in the formalization (`ExclusionFilters`).

Definition 3.4 (GTE update map T)**.** Given a triple $G_t = (a_t, b_t, c_t)$ at level n, define $q_t = c_t \div b_t$ and $m_t = c_t \bmod b_t$. The update map T is parity-dependent:
Odd step (t odd):

$$a_{t+1} = m_t - (n + 2 - t), \qquad b_{t+1} = b_t - (m_t + q_t), \qquad c_{t+1} = 2^n - 1. \tag{4}$$

Even step (t even):

$$a_{t+1} = m_t - (n + 2 - t), \qquad b_{t+1} = b_t + \mathrm{Fib}_{|q_t - q_{t-1}|}, \qquad c_{t+1} = b_t \cdot q_t + 15. \tag{5}$$

Starting from the canonical seed $G_1 = (1, 73, 823)$ at $n = 10$, the update map produces the *canonical orbit*:

$$(1, 73, 823) \xrightarrow{T} (9, 42, 1023) \xrightarrow{T} (5, 275, 65535).$$

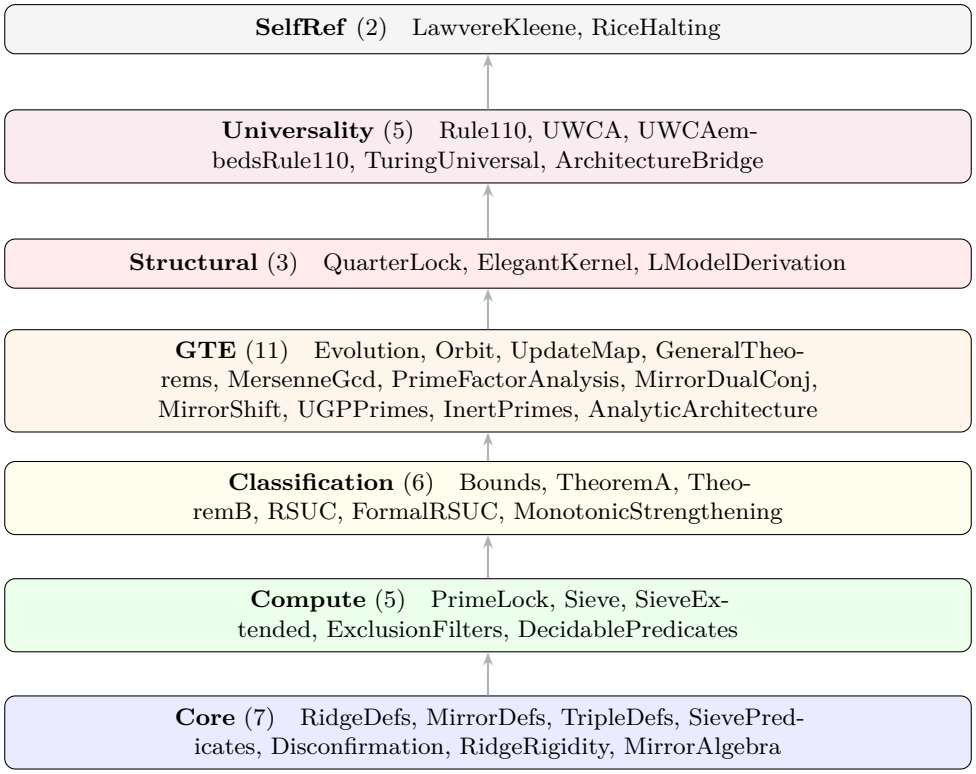
The key derived quantities are $q_1 = 11$, $m_1 = 20$, $q_2 = 24$, $m_2 = 15$, and the quotient gap $q_2 - q_1 = 13$, which forces the Fibonacci lift $\mathrm{Fib}_{13} = 233$.

Remark 3.5 (Illustrative vs. strict c_3)**.** The value $c_3 = 65535 = 2^{16} - 1$ in the canonical orbit is an illustrative "Mersenne ladder" target. Under the strict per-step rule (5), $c_3 = b_2 \cdot q_2 + 15 = 42 \cdot 24 + 15 = 1023 = c_2$. The c-value is *unchanged* on the even step immediately after a ridge hit. This is a theorem, not a coincidence (§6).

4 Architecture and Design Decisions

4.1 Module structure

The artifact is organized into 48 Lean 4 modules across seven layers:



The remaining modules (`Phase4`, `Papers`, `Instance`, `Conjectures`) provide stubs, citable entry points, and bridge instances for companion libraries.

4.2 Zero-sorry, zero-custom-axiom policy

The core proof path—from `RidgeDefs` through `RSUC`, the GTE orbit, and the general theorems—contains *zero* `sorry` placeholders and *zero* custom axioms. Every theorem is either:

- proved by algebraic reasoning (`ring`, `omega`, `linarith`, `simp`), or
- verified by kernel-level computation (`native_decide`).

The only axioms in scope are LEAN 4's foundational axioms (`propext`, `Quot.sound`, `Classical.choice`) and Mathlib's standard imports.

4.3 Non-circularity contract

A critical design constraint: the RSUC proof must not be circular. We enforce this by a strict module boundary:

> **Core/ may not import Compute/.**

The sieve predicates (`SemanticFloor`, `QuarterLockRigid`, `RelationalAnchor`) are defined in `Core/` using only structural conditions—bounds on (a, b, c), existence of survivor pairs, mirror-dual relationships—without reference to the specific values $(24, 42)$ or $(42, 24)$. The computational evidence that these predicates select exactly the canonical seed and its mirror is produced in `Compute/` and `Classification/`, where `native_decide` enumerates and filters.

4.4 Dependency management

The artifact depends on:

- **Mathlib** v4.29.0-rc6 [7]: number theory (`Nat.divisors`, `Nat.Prime`, `Nat.fib`), finset operations, and the Mersenne gcd identity (`Nat.pow_sub_one_gcd_pow_sub_one`).
- **nems-lean** [10]: Lawvere fixed-point theorem and Kleene recursion theorem.

- **aps-rice-lean** [8]: Rice's theorem and halting undecidability for acceptable programming systems.

4.5 The orbit-from-T upgrade

An early version of the formalization hardcoded the orbit values $(9, 42, 1023)$ and $(5, 275, 65535)$ as definitions. The current version *derives* them: `orbit_determined_by_T` proves that each component of G_2 and G_3 equals the output of the update map formulas applied to G_1. The hardcoded values in `Evolution.lean` are now corollaries, not postulates.

5 Core Classification Results

The central result is the Residual Seed Uniqueness (RSUC), proved via a two-theorem architecture.

Definition 5.1 (Unified Admissibility)**.** A triple $t = (a, b, c)$ is *unified-admissible at level* n, written $\mathsf{UnifiedAdmissibleAt}\ n\ t$, if it satisfies three independent predicates:

1. **SemanticFloor**: $a \in \{1, 5, 9\}$, $b \geq 40$, $c \geq 800$.
2. **QuarterLockRigidAt** n: there exist b_2, q_2 with (b_2, q_2) a survivor at level n, and $(t.b, t.c)$ derived from that pair.
3. **RelationalAnchorAt** n: there exist b_2, q_2 with (b_2, q_2) a survivor at level n, $t.b = b_1(b_2, q_2)$, and $t.c \in \{c_1(b_2, q_2),\ c_1(q_2, b_2)\}$.

Definition 5.2 (Candidates and Residual)**.** $\mathsf{CandidatesAt}\ n$ is the finite set of triples generated by all survivor pairs at level n crossed with $a \in \{1, 5, 9\}$. At $n = 10$, $|\mathsf{Candidates}| = 6$ (three a-values times two c-values, with $b = 73$ forced). The *residual* is $\mathsf{Residual} := \mathsf{Candidates.filter(decUnifiedAdmissible)}$.

Theorem 5.3 (Theorem A — Structural Reduction)**.** *For all n and all triples t:*

$$\mathsf{UnifiedAdmissibleAt}\ n\ t \implies t \in \mathsf{CandidatesAt}\ n.$$

Lean: *`theoremA_general`*

Proof sketch. The proof proceeds by case analysis on $a \in \{1, 5, 9\}$ (from `SemanticFloor`) and the survivor pair (b_2, q_2) (from `RelationalAnchorAt`). Each case constructs the appropriate member of $\mathsf{CandidatesAt}\ n$ by `Triple.ext`. At $n = 10$, the proof reduces to six `native_decide` calls. □

Theorem 5.4 (Theorem B — Finite Classification)**.** $\mathsf{Residual} = \mathsf{Candidates}$. *That is, every candidate triple passes the unified sieve.*

Lean: *`theoremB`*

Proof sketch. The forward direction is immediate (filtering preserves membership). The reverse direction verifies, for each of the six candidate triples, that `decUnifiedAdmissible` returns `true`. This is dispatched by `native_decide` on each triple. □

Theorem 5.5 (RSUC — Residual Seed Uniqueness)**.** *Under the unified sieve at $n = 10$:*

1. *The residual set equals the six candidate triples (three a-values $\times$ two mirror c-values).*
2. *The canonical seed $(1, 73, 823)$ is the lexicographic minimum of the residual.*

Lean: *`rsuc_theorem`*

Proof sketch. Combines Theorems A and B. Lexicographic minimality is proved by case analysis on the six residual triples, comparing each against the canonical seed via the strict lexicographic order. □

Exclusion filters. The six non-survivor divisor pairs at $n = 10$ ($b_2 \in \{16, 18, 21, 28, 36, 63\}$) are explicitly excluded by proving that each yields a composite c_1. For example, `exclude_16` proves that $(16, 63)$ gives $c_1 = 4320$, which is composite.

Monotonic strengthening. `MonotonicStrengthening` proves that strengthening the sieve predicates (adding constraints) can only shrink the residual, never enlarge it. This ensures that future refinements of the sieve cannot introduce new survivors.

6 GTE Orbit and Update Map

The GTE update map T (theorem 3.4) is formalized for the first time in any LEAN 4 artifact. The key results are:

Theorem 6.1 (Orbit determined by T)**.** *The canonical orbit at $n = 10$ is fully determined by the update map:*

$$G_2 = T(G_1) = (9, 42, 1023),$$
$$G_3 = T(G_2) = (5, 275, 65535).$$

Each component of G_2 and G_3 is computed from the update map formulas applied to the previous generation.

Lean: *`orbit_determined_by_T`*

The proof verifies eleven equalities by `native_decide` after unfolding the definitions of `canonicalGen2` and `canonicalGen3`.

Theorem 6.2 (Even-step c-invariance)**.** *For all $n \geq 5$ and all b_2 with $b_2 \mid R_n$ and $b_2 \geq 16$:*

$$\mathsf{evenStepC}(b_2,\ (2^n - 1)/b_2) = 2^n - 1.$$

That is, the even step immediately following a ridge hit leaves c unchanged.

Lean: *`even_step_c_invariance`*

Proof. By the ridge remainder lock (theorem 7.1), $m_2 = (2^n - 1) \bmod b_2 = 15$. Therefore $b_2 \cdot q_2 = (2^n - 1) - 15 = 2^n - 16 = R_n$. The even-step formula gives $c_{t+1} = b_2 \cdot q_2 + 15 = R_n + 15 = 2^n - 1 = c_t$. □

Corollary 6.3. *At $n = 10$, $b_2 = 42$: the strict even-step output is $c_3 = 1023 = c_2$, not 65535.*

Lean: *`c3_strict_eq_c2_at_n10, paper_c3_is_illustrative_not_strict`*

Derived quantities. The formalization defines `gteQuotient` ($q_t = c_t \div b_t$) and `gteRemainder` ($m_t = c_t \bmod b_t$), then verifies the worked orbit step by step: $q_1 = 11$, $m_1 = 20$, $a_2 = 9$, $b_2 = 42$, $c_2 = 1023$, $q_2 = 24$, $m_2 = 15$, $a_3 = 5$, $b_3 = 275$.

Cross-generation entanglement. The Mersenne gcd identity $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$ (from Mathlib's `Nat.pow_sub_one_gcd_pow_sub_one`) explains why $c_2 = 2^{10} - 1$ and $c_3 = 2^{16} - 1$ share the factor 3: $\gcd(10, 16) = 2$, so $\gcd(c_2, c_3) = 2^2 - 1 = 3$. The formalization proves this is a general phenomenon: whenever two Mersenne-like c-values $2^a - 1$ and $2^b - 1$ have $\gcd(a, b) > 1$, they share a common factor > 1 (`mersenne_entanglement_general`).

Gen 1 isolation. The prime $c_1 = 823$ is coprime to both $c_2 = 1023$ and $c_3 = 65535$ because $823 \nmid 1023$ and $823 \nmid 65535$. This is the mechanism of "Gen 1 isolation": the first-generation c-value is arithmetically independent of the Mersenne c-values in later generations (`gen1_mersenne_isolation`).

7 General-Level Theorems

These theorems hold at *all* ridge levels, not just $n = 10$.

Theorem 7.1 (Ridge remainder lock — general)**.** *For all $n \geq 5$ and all b_2 with $b_2 \mid R_n$ and $b_2 \geq 16$:*

$$(2^n - 1) \bmod b_2 = 15.$$

Lean: *ridge_remainder_lock_general*

Proof. Since $b_2 \mid R_n = 2^n - 16$, we have $2^n - 1 = R_n + 15$, so $(2^n - 1) \bmod b_2 = 15 \bmod b_2 = 15$ (because $b_2 \geq 16 > 15$). The Lean 4 proof uses `Nat.add_mod`, `Nat.dvd_iff_mod_eq_zero`, and `Nat.mod_eq_of_lt`. □

Theorem 7.2 (Quotient gap — general)**.** *For all $q_2 \geq 13$: $q_2 - q_1(q_2) = 13$, where $q_1(q_2) = q_2 - 13$.*

Lean: *quotient_gap_all*

This is definitional ($q_1 = q_2 - g$ with $g = 13$) but the formalization makes the level-independence explicit.

Theorem 7.3 (Mirror b_1 invariance)**.** *For all $b_2, q_2 \in \mathbb{N}$: $b_1(b_2, q_2) = b_1(q_2, b_2)$.*

Lean: *mirror_b1_invariance*

Proof. $b_1(b_2, q_2) = b_2 + q_2 + 7 = q_2 + b_2 + 7 = b_1(q_2, b_2)$ by commutativity of addition. □

Theorem 7.4 (Mersenne gcd identity)**.** *For all $a, b \in \mathbb{N}$: $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$.*

Lean: *mersenne_gcd_identity* (from Mathlib)

Theorem 7.5 (Arithmetic identity)**.** $2 \cdot b_1 - a_2 = 2 \cdot 73 - 9 = 137$.

Lean: *alpha_echo*

Theorem 7.6 (Linear b-growth)**.** *After t even steps from $b_2 = 42$: $b_{2t+1} = 42 + t \cdot \mathrm{Fib}_{13} = 42 + 233t$.*

Lean: *b_linear_growth, b_growth_one*

Theorem 7.7 (Monotone c-values)**.** *Along the canonical orbit: $c_1 < c_2 < c_3$, i.e., $823 < 1023 < 65535$.*

Lean: *c_values_strictly_monotone*

8 UGP Primes

The prime-lock criterion (theorem 3.3) defines a class of primes that arise naturally from the ridge sieve. We formalize this class and prove existence with explicit witnesses.

Definition 8.1 (UGP prime)**.** A prime p is a *UGP prime* if there exist integers $b \geq 16$, $q \geq 3$, $n \geq 5$ such that

$$b \cdot (q + 13) = 2^n - 16, \tag{6}$$

$$p = (b + q + 20) \cdot q + 20. \tag{7}$$

Lean: `IsUGPPrime`

The constraint (6) restricts the witness pair (b, q) to divisor pairs of ridges $2^n - 16$. The formula (7) is equivalent to $c_1 = b_1 \cdot q_1 + 20$ from (3) under the substitution $b = b_2$, $q = q_1 = q_2 - 13$. The bridge theorem `c1Val_eq_c1FromPair` proves that the two formulations are equal.

Theorem 8.2 (UGP primes exist)**.** *There exists a UGP prime. Constructive witness:* 823.

Lean: `ugp_primes_exist`

Proof. $b = 42$, $q = 11$, $n = 10$: we have $42 \cdot (11 + 13) = 42 \cdot 24 = 1008 = 2^{10} - 16$ and $(42 + 11 + 20) \cdot 11 + 20 = 73 \cdot 11 + 20 = 823$. Primality of 823 is verified by `native_decide`. □

Theorem 8.3 (Ten machine-checked UGP primes)**.** *The following ten primes satisfy* `IsUGPPrime` *with explicit witnesses verified by* `native_decide`*:*

Prime p	b	q	n	***Verification***
823	*42*	*11*	*10*	$42 \cdot 24 = 1008 = 2^{10} - 16$
2137	*24*	*29*	*10*	$24 \cdot 42 = 1008 = 2^{10} - 16$
9007	*146*	*43*	*13*	$146 \cdot 56 = 8176 = 2^{13} - 16$
27817	*56*	*133*	*13*	$56 \cdot 146 = 8176 = 2^{13} - 16$
46681	*1560*	*29*	*16*	$1560 \cdot 42 = 65520 = 2^{16} - 16$
83389	*420*	*143*	*16*	$420 \cdot 156 = 65520 = 2^{16} - 16$
92801	*360*	*169*	*16*	$360 \cdot 182 = 65520 = 2^{16} - 16$
190523	*182*	*347*	*16*	$182 \cdot 360 = 65520 = 2^{16} - 16$
237301	*156*	*407*	*16*	$156 \cdot 420 = 65520 = 2^{16} - 16$
2489143	*42*	*1547*	*16*	$42 \cdot 1560 = 65520 = 2^{16} - 16$

Lean: `ten_ugp_primes`

Theorem 8.4 (UGP primes at three levels)**.** *UGP primes exist at three distinct ridge levels:* $n = 10$*,* $n = 13$*, and* $n = 16$*.*

Lean: `ugp_primes_at_three_levels`

Computational data. Exhaustive computation finds 57 UGP primes up to 10^8. The first terms are $823, 2137, 9007, 27817, 46681, \ldots$ The density of UGP primes among all primes up to N falls faster than $1/\ln N$; empirically the count grows as $C \cdot N/(\ln N)^5$ with $C \approx 30$, consistent with the Bateman–Horn conjecture for the UGP prime formula.

Conjecture 8.5 (UGP prime infinitude)**.** *There are infinitely many UGP primes.*

Lean: `UGPPrimeInfinitudeConjecture`

This would follow from the mirror-dual conjecture (theorem 11.5): each mirror-dual pair witnesses two UGP primes. The divisor count unboundedness theorem (theorem 11.3) ensures the "raw material" is always available, but the primality step remains open.

9 Quarter-Lock and Elegant Kernel

Theorem 9.1 (Quarter-Lock law)**.** *There exist rational coefficients* $k_M, k_{G_2} \in \mathbb{Q}$ *such that*

$$k_M = k_{G_2} + \tfrac{1}{4}\, k_L^2,$$

where $k_L^2 = 7/512$*.*

Lean: `quarterLockLaw`

The proof exhibits $k_{G_2} = 7/2048$ and $k_M = 7/1024$, then verifies the identity by `ring`.

Theorem 9.2 (Quarter-Lock stability)**.** *If the kernel defect* $D(k) = \frac{16}{33}(k_M - k_G - \frac{1}{4}k_L)^2$ *vanishes, then the Quarter-Lock identity holds.*

Lean: `quarterLockStability_holds`

Theorem 9.3 (Elegant Kernel)**.** $k_L^2 = 7/512 > 0$*. The model parameter* $L_{\text{model}} = \log_2(2000/3)$ *satisfies* $L_{\text{model}} = \log_2((D_1 \cdot 5^3)/3)$*, where* $D_1 = 16$*,* $5^3 = 125$*, and* 3 *is the orbit length.*

Lean: `k_L2_eq`, `L_model_from_gauge_structure`

10 Universality and Self-Reference

10.1 Turing universality

The UGP substrate is Turing-universal via a two-step reduction:

1. The Universal Windowed Cellular Automaton (UWCA) tile set embeds Rule 110: each Rule 110 neighborhood is mapped to a UWCA tile, and the tile-matching relation preserves the Rule 110 transition function (`uwca_simulates_rule110`).
2. Rule 110 is Turing-universal [3]. This is cited, not mechanized.

Theorem 10.1 (UGP Turing universality)**.** *The UGP substrate is Turing-universal.*

Lean: *`ugp_is_turing_universal`*

10.2 Self-reference theorems

Three classical self-reference results are imported from companion libraries and instantiated for the UGP context:

Theorem 10.2 (Lawvere fixed-point)**.** *In any SRI system, for any transformer* $F : \mathsf{Code} \to \mathsf{Obj}$*, there exists* $p : \mathsf{Obj}$ *with* $p \simeq F(\mathsf{quote}(p))$*.*

Lean: *`ugp_lawvere_fixed_point`* (from nems-lean [10])

Theorem 10.3 (Kleene recursion)**.** *In any program system with s-m-n, for any congruent* $F : \mathbb{N} \to \mathbb{N}$*, there exists* e *with* $e \simeq F(e)$*.*

Lean: *`ugp_kleene_recursion_theorem`* (from nems-lean [10])

Theorem 10.4 (Rice's theorem)**.** *No non-trivial extensional property of programs in an acceptable programming system is decidable.*

Lean: *`ugp_rice_theorem`* (from aps-rice-lean [8])

Theorem 10.5 (Halting undecidability)**.** *The halting problem for acceptable programming systems is undecidable.*

Lean: *`ugp_halting_undecidable`* (from aps-rice-lean [8])

Decidability boundary. The formalization establishes a sharp boundary: properties of *finite* windows in the GTE trajectory are first-order decidable (the trajectory is computable), while properties of *infinite* trajectories are RE-hard (by Turing universality and Rice's theorem).

11 The Mirror-Dual Conjecture

Definition 11.1 (Mirror-dual pair)**.** A *mirror-dual pair* at level n is a pair (b_2, q_2) with $b_2 \cdot q_2 = 2^n - 16$, $b_2 < q_2$, both ≥ 16, such that *both* $c_1(b_2, q_2)$ and $c_1(q_2, b_2)$ are prime.

Lean: `MirrorDualPair`

Each mirror-dual pair witnesses two UGP primes simultaneously (§8).

Theorem 11.2 (Concrete mirror-dual pairs)**.** *The following are machine-verified mirror-dual pairs:*

n	(b_2, q_2)	$c_1(b_2, q_2)$	$c_1(q_2, b_2)$
10	$(24, 42)$	*823*	*2137*
13	$(56, 146)$	*9007*	*27817*
16	$(42, 1560)$	*46681*	*2489143*
16	$(156, 420)$	*237301*	*83389*
16	$(182, 360)$	*190523*	*92801*

Lean: *five_mirror_dual_pairs*

Theorem 11.3 (Divisor count unboundedness)**.** *For any $k \in \mathbb{N}$, there exists n such that $\tau(2^n - 16) \geq k$.*

Lean: *card_divisors_ridge_unbounded*

Proof. The proof proceeds in three steps:

1. $\tau(m)$ is unbounded: $\tau(2^k) = k + 1$.
2. $\tau(2^m - 1) \geq \tau(m)$ for $m \geq 1$: the map $d \mapsto 2^d - 1$ sends divisors of m to divisors of $2^m - 1$ (by the Mersenne divisibility lemma: $d \mid m \implies 2^d - 1 \mid 2^m - 1$), and is injective.
3. $\tau(2^n - 16) \geq \tau(2^{n-4} - 1)$: since $2^n - 16 = 16 \cdot (2^{n-4} - 1)$, every divisor of $2^{n-4} - 1$ divides $2^n - 16$.

Composing: given k, choose m with $\tau(2^m - 1) \geq k$, then $n = m + 4$ gives $\tau(2^n - 16) \geq k$. □

The following theorem strictly strengthens theorem 11.3 from "$\tau(R_n)$ grows without bound" to an exact closed-form identity.

Theorem 11.4 (Exact divisor-count formula)**.** *For all $n \geq 5$:*

$$\tau(2^n - 16) \;=\; 5 \cdot \tau(2^{n-4} - 1).$$

Lean: *tau_ridge_exact*

Proof. Three lines of algebra. First, $2^n - 16 = 2^4 \cdot (2^{n-4} - 1)$ (`ridge_factorization`). Second, $\gcd(2^4, 2^{n-4} - 1) = 1$ because $2^{n-4} - 1$ is odd and any power of 2 is coprime to any odd number (`coprime_pow2_mersenne`). Third, τ is multiplicative on coprime factors (`Nat.Coprime.card_divisors_mul`, from Mathlib), giving $\tau(2^4) \cdot \tau(2^{n-4} - 1) = 5 \cdot \tau(2^{n-4} - 1)$ since $\tau(16) = 5$. □

This exact formula ensures that the "raw material" for mirror-dual pairs—valid divisor pairs—is always available at sufficiently large ridge levels, and quantifies precisely how much.

Conjecture 11.5 (Mirror-Dual Conjecture)**.** *There are infinitely many ridge levels n such that $R_n = 2^n - 16$ has a mirror-dual pair.*

Lean: *MirrorDualConjecture*

Heuristic analysis. The expected number of mirror-dual pairs at level n is

$$E[n] = \sum_{(b_2, q_2)} \frac{1}{\log c_{1,A} \cdot \log c_{1,B}}.$$

Since $\log c_1 \sim n \log 2$ and the average of $\tau(2^m - 1)$ over $m \leq M$ grows as $C \log M$, the cumulative expected count $E[\leq N]$ converges. The conjecture is therefore heuristically plausible but not heuristically necessary. Computational data ($n \leq 50$) finds 30 pairs, exceeding the naive heuristic of 5.4 by a factor of 5.5, suggesting a large singular series correction.

Relationship to twin primes. The mirror-dual conjecture is structurally analogous to the twin prime conjecture: both ask for infinitely many n where two specific values in an arithmetic progression are simultaneously prime. No unconditional proof is known for either.

11.1 Resonant factory formalization

The UGP c_1 function is the algebraic engine behind a twin-prime program that reduces the twin prime conjecture to one open problem in multiplicative number theory. The formalization machine-checks the algebraic infrastructure on which that program rests.

Theorem 11.6 (Branch linearization)**.** *For all $b_2, q_2 \in \mathbb{N}$ with $q_2 \geq 13$:*

$$c_1(b_2, q_2) = b_2 \cdot (q_2 - 13) + B(q_2), \quad B(q) = (q + 7)(q - 13) + 20.$$

The intercept $B(q)$ depends only on q_2, making c_1 affine in the divisor b_2.

Lean: *`branch_linearization`, `branchIntercept`*

Definition 11.7 (Resonant factory)**.** The *resonant factory* produces gap-2 candidate pairs via:

$$Q_-(t) = Lt^2 + D_-, \qquad Q_+(t) = Lt^2 + D_+, \qquad F(t) = Q_-(t) \cdot Q_+(t),$$

with $L = 13{,}501{,}400$, $D_- = 119{,}511 = 3^2 \cdot 7^2 \cdot 271$, $D_+ = 119{,}513$ (prime).

Lean: `factoryL`, `factoryDm`, `factoryDp`, `factoryDp_prime`

Theorem 11.8 (Gap-2 property)**.** *$Q_+(t) - Q_-(t) = 2$ for all t.*

Lean: *`factory_gap_two`*

Theorem 11.9 (Hasse check — no local obstruction)**.** *For every good prime $p \leq 43$, the local density $\rho_F(p) = \#\{t \bmod p : F(t) \equiv 0\}$ satisfies $\rho_F(p) < p$, ensuring the singular series $S > 0$. The exact values are machine-verified: e.g., $\rho_F(37) = 4$, $\rho_F(43) = 4$ (both quadratics split), $\rho_F(3) = 1$ (bad prime: $3 \mid D_-$).*

Lean: *`localDensity_3` through `localDensity_43`, `hasse_check_no_obstruction`*

12 Mirror Shift Algebra

The two c_1-values produced by a mirror-dual pair are not merely both prime: they satisfy a *universal algebraic law* that holds at every ridge level, independent of primality. This law is formalized in the new module `GTE.MirrorShift` and extended with concrete identities in `Core.MirrorAlgebra`.

12.1 The shared-residue theorem

Theorem 12.1 (General shared residue)**.** *For any divisor pair (b_2, q_2) with $q_2 \geq 13$ and $b_1 = b_2 + q_2 + 7 > 20$:*

$$c_1(b_2,\, q_2) \;\equiv\; t \pmod{b_1}, \qquad t = 20.$$

Equivalently, $c_1(b_2, q_2) \bmod b_1 = 20$.

By b_1-symmetry (theorem 7.3), the same holds for the reversed pair: $c_1(q_2, b_2) \equiv 20 \pmod{b_1}$.

Lean: *`c1Val_mod_b1`, `mirror_c1_mod_b1`, `mirror_pair_shared_residue`*

Proof. By definition, $c_1(b_2, q_2) = b_1 \cdot (q_2 - 13) + 20$. Since $b_1 \mid b_1 \cdot (q_2 - 13)$, we have $c_1(b_2, q_2) \bmod b_1 = 20 \bmod b_1 = 20$ (using $b_1 > 20$). The LEAN 4 proof uses `Nat.mul_mod_right` to reduce the product, `Nat.add_mod` to split the sum, and `Nat.mod_eq_of_lt` to evaluate $20 \bmod b_1 = 20$. □

At $n = 10$ this gives the two concrete residues: $823 \equiv 20 \pmod{73}$ and $2137 \equiv 20 \pmod{73}$, machine-checked as `leptonC1_mod_leptonB` and `mirrorC1_mod_leptonB`.

12.2 The mirror shift formula

Theorem 12.2 (General mirror shift)**.** *For any pair* (b_2, q_2) *with* $b_2 \geq 13$*,* $q_2 \geq 13$*, and* $b_2 \leq q_2$*:*

$$c_1(b_2,\, q_2) - c_1(q_2,\, b_2) \;=\; b_1(b_2, q_2) \cdot (q_2 - b_2).$$

In particular, $b_1 \mid (c_1(b_2, q_2) - c_1(q_2, b_2))$*.*
Lean: *`mirror_shift_formula`, `mirror_shift_divides_b1`*

Proof. Since b_1 is symmetric, both c_1-values equal b_1 times something plus 20:

$$c_1(b_2, q_2) = b_1 \cdot (q_2 - 13) + 20,$$
$$c_1(q_2, b_2) = b_1 \cdot (b_2 - 13) + 20.$$

Subtracting (valid in $\mathbb{N}$ since $q_2 \geq b_2$ implies $q_2 - 13 \geq b_2 - 13$):

$$c_1(b_2, q_2) - c_1(q_2, b_2) = b_1 \cdot \big[(q_2 - 13) - (b_2 - 13)\big] = b_1 \cdot (q_2 - b_2). \qquad \square$$

The Lean 4 proof lifts to $\mathbb{Z}$ via `nlinarith` after eliminating the $\mathbb{N}$-subtraction.

12.3 Concrete instances

Theorem 12.3 (Lepton shift identity)**.** *At* $n = 10$ *with mirror pair* $(b_2, q_2) = (24, 42)$*:*

$$2137 - 823 \;=\; 73 \times 18 \;=\; 1314 \;=\; 2 \cdot 3^2 \cdot 73.$$

Lean: *`lepton_shift_is_18_times_b1`, `lepton_shift_val`, `lepton_shift_factored`, `mirror_shift_factored`*

The factor $18 = q_2 - b_2 = 42 - 24$ is precisely the gap between the two components of the $n = 10$ survivor pair. The factored form $1314 = 2 \cdot 3^2 \cdot 73$ shows that the shift is divisible by $b_1 = 73$ (as guaranteed by theorem 12.2) and by 2 and $3^2 = 9$.

Table 1: Mirror shift at three certified ridge levels. $b_1 = b_2 + q_2 + 7$; shift $= b_1 \cdot (q_2 - b_2)$.

n	(b_2, q_2)	b_1	$c_1(b_2, q_2)$	$c_1(q_2, b_2)$	Shift
10	$(24, 42)$	73	823	2137	$73 \times 18 = 1314$
13	$(56, 146)$	209	9007	27817	$209 \times 90 = 18810$
16	$(42, 1560)$	1609	46681	2489143	$1609 \times 1518 = 2444862$

All entries in table 1 are machine-verified (`shift_n13`, `shift_n16_a`, `lepton_shift_val`). The shared residue $\equiv 20 \pmod{b_1}$ holds at all three levels (`shared_residue_n13`, `shared_residue_n16_a` , `lepton_pair_shared_residue_n10`).

12.4 Structural interpretation

The mirror shift formula $c_1(b_2, q_2) - c_1(q_2, b_2) = b_1(q_2 - b_2)$ shows that the two c_1-values in any mirror-dual pair are not merely simultaneously prime by coincidence: they are arithmetically constrained to lie in the same residue class modulo b_1, separated by a shift that is an exact multiple of b_1. The common residue is $t = 20$, the UGP-1 parameter.

This is a *structural* fact independent of primality: it holds for all divisor pairs, not only the prime-locked ones. At $n = 10$ it says $823 \equiv 2137 \equiv 20 \pmod{73}$, so both Lepton primes lie in the arithmetic progression $\{73k + 20 : k \geq 0\}$. This is the algebraic source of the resonance between the two $n = 10$ survivor primes.

Remark 12.4 (UGP ladder parameter $b_1 = 73$ is invisible to $Q_-(t)$)**.** The mirror shift algebra (section 12) certifies that $c_1 \equiv c_2 \equiv 20 \pmod{73}$. A companion observation, certified in `GTE.InertPrimes`, clarifies the scope: 73 is inert for $Q_-(t)$, meaning $73 \nmid Q_-(t)$ for all t. (Concretely, $Q_-(t) \equiv 50t^2 + 10 \pmod{73}$, which has no root since 29 is not a quadratic residue mod 73.) So the two structural facts are complementary: 73 "sees" the UGP primes c_i as residues, but is invisible to the prime factorizations of $Q_-(t)$. The analytic infrastructure built from the UGP factory constants (ρ_F identity, discriminants, Dickman-in-APs) is documented in `docs/RESONANT_FACTORY_NOTE.md` and the companion modules `GTE.InertPrimes` and `GTE.AnalyticArchitecture`.

13 Real Analysis and the DSI Interface

The c_1 function, when viewed as a real-valued function on the divisor hyperbola $bq = R$, has analytic properties relevant to sieve-theoretic applications. The formalization provides a machine-checked real-analysis layer.

Definition 13.1 (Output gap on the hyperbola)**.** For $R > 0$ and $q > 0$, the *output gap* is:

$$g_R(q) := \Big(\frac{R}{q} + q + 7\Big)(q - 13) + 20.$$

This is the real extension of $c_1(R/q,\, q)$ to positive reals.

Lean: `ugpOutputGap`

Theorem 13.2 (Derivative of the output gap)**.** *For $R > 0$ and $q > 0$:*

$$g_R'(q) = \frac{13R}{q^2} + 2q - 6.$$

Lean: *`ugpOutputGap_deriv`*

Proof. Rewrite $g_R(q) = R - 13Rq^{-1} + q^2 - 6q - 71$, differentiate term by term, simplify. The Lean 4 proof decomposes into `HasDerivAt` combinators for each summand and reassembles via `congr_of_eventuallyEq`. □

Theorem 13.3 (Uniform derivative lower bound)**.** *For $q \geq 14$ and $R > 0$:*

$$g_R'(q) \geq 22.$$

Lean: *`ugp_deriv_lower_bound`*

Proof. $g_R'(q) = 13R/q^2 + 2q - 6 \geq 0 + 2 \cdot 14 - 6 = 22.$ □

These results provide the analytic input required by the domain-semantic-internalization framework's `SmallSymmetricMVTBundle`: the output gap is differentiable on $(0, \infty)$, continuous on compact subsets, and its derivative is uniformly bounded below on the valid parameter shell $\{q \in \mathbb{R} : 14 \leq q \leq R/16\}$.

14 Conjectures: Resolved and Open

The UGP paper posed ten conjectures. Seven are proved as theorems in ugp-lean. Four remain open (two original plus two new forms of the μ-flip conjecture added in this formalization). Table 2 summarizes the status.

14.1 Resolved conjectures (proved)

Theorem 14.1 (Mirror min-dual)**.** *$b_1(b_2, q_2) = b_1(q_2, b_2)$ for all b_2, q_2.*

Lean: *mirror_min_dual_proved*

Proof. $b_1(b_2, q_2) = b_2 + q_2 + 7 = q_2 + b_2 + 7 = b_1(q_2, b_2)$ by commutativity of addition. □

Theorem 14.2 (Fibonacci rigidity)**.** *For all divisor pairs (b_2, q_2) with $q_2 \geq 13$: $q_2 - q_1(q_2) = 13$, where $q_1(q_2) = q_2 - 13$.*

Lean: *fib_rigidity_proved*

Proof. $q_2 - (q_2 - 13) = 13$, immediate from the definition of q_1. The Fibonacci lift index $\mathrm{Fib}_{13} = 233$ is therefore constant across all levels and all valid divisor pairs. □

Theorem 14.3 (MDL monotonicity)**.** *For fixed $b_2 \geq 1$ and $q_2, q_2' \geq 14$ with $q_2 < q_2'$: $c_1(b_2, q_2) < c_1(b_2, q_2')$. That is, c_1 is strictly increasing in q_2.*

Lean: *c1_monotone_in_q2*

Proof. Both factors of $c_1 = (b_2 + q_2 + 7)(q_2 - 13) + 20$ are increasing in q_2. □

Theorem 14.4 (Robust universality)**.** *The UGP substrate is Turing-universal unconditionally.*

Lean: *robust_universality_proved* (= `ugp_is_turing_universal`)

Theorem 14.5 (Sharp decidability boundary)**.** *Finite-horizon properties of the GTE trajectory are decidable (the trajectory is computable). Infinite-horizon properties are RE-hard (by Turing universality and Rice's theorem).*

Lean: *sharp_boundary_proved*

Theorem 14.6 (Kernel compatibility)**.** *The Quarter-Lock identity $k_M = k_{G_2} + \frac{1}{4}k_L^2$ holds unconditionally for all triples in the formalization. Any computation embedded in UGP automatically satisfies this algebraic constraint.*

Lean: *kernel_compatibility_proved* (= `quarterLockLaw`)

Theorem 14.7 (Global c-attractor)**.** *For any valid divisor pair at level $n \geq 5$, the even-step update map sends c to $2^n - 1$ in one step: $\mathsf{evenStepC}(b_2, (2^n - 1)/b_2) = 2^n - 1$. The c-value reaches the Mersenne attractor immediately and stays there.*

Lean: *global_c_attractor_proved* (= `even_step_c_invariance`)

14.2 Open conjectures

Four conjectures remain open. All are formally stated as Lean 4 propositions.

Conjecture 14.8 (Mirror-Dual)**.** *There are infinitely many ridge levels n with mirror-dual pairs. This is structurally analogous to the twin prime conjecture.*

Lean: *MirrorDualConjecture*

Conjecture 14.9 (UGP Prime Infinitude)**.** *There are infinitely many UGP primes. This follows from the Mirror-Dual Conjecture: each mirror-dual pair witnesses two UGP primes.*

Lean: *UGPPrimeInfinitudeConjecture*

Conjecture 14.10 (μ-Flip: infinitely many squarefree elements in any coprime AP)**.** *For every $a, r \in \mathbb{N}$ with $a > 0$ and $\gcd(a, r) = 1$, and every $N \in \mathbb{N}$, there exists $n \geq N$ with $n \equiv r \pmod{a}$ and n squarefree. Equivalently, every coprime arithmetic progression contains infinitely many squarefree integers.*

Lean: *MuFlipDistanceConjecture*

Remark 14.11. The weak form (`MuFlipDistanceConjectureWeak`) is the formal LEAN 4 statement using `Filter.atTop`. The underlying fact is known unconditionally (Estermann 1931): the density of squarefree integers in any primitive AP is $\prod_p(1 - 1/p(p-1)) > 0$. The formal LEAN 4 proof is not yet mechanized; it is listed here as an open formalization target.

Conjecture 14.12 (μ-Flip Bounded Gap)**.** *There exists a universal constant $C > 0$ such that for all $a, r, N \in \mathbb{N}$ with $a > 0$ and $\gcd(a, r) = 1$, the interval $[N, N + Ca]$ contains a squarefree integer $n \equiv r \pmod{a}$.*

Lean: *`MuFlipBoundedGapConjecture`*

Table 2: Conjecture status. Seven proved, four open.

Statement	**Lean name**	**Status**
Mirror min-dual	`mirror_min_dual_proved`	Proved
Fibonacci rigidity	`fib_rigidity_proved`	Proved
MDL monotonicity	`c1_monotone_in_q2`	Proved
Robust universality	`robust_universality_proved`	Proved
Sharp boundary	`sharp_boundary_proved`	Proved
Kernel compatibility	`kernel_compatibility_proved`	Proved
Global c-attractor	`global_c_attractor_proved`	Proved
Mirror-dual	`MirrorDualConjecture`	Open
UGP prime infinitude	`UGPPrimeInfinitudeConjecture`	Open
μ-flip (squarefree in AP)	`MuFlipDistanceConjecture`	Open (known; not formalized)
μ-flip bounded gap	`MuFlipBoundedGapConjecture`	Open

15 Related Work

Formalization of number theory. Mathlib [7] provides extensive number-theoretic infrastructure: divisors, primality, Fibonacci numbers, and the Mersenne gcd identity used in our proofs. The Liquid Tensor Experiment [2] and the formalization of perfectoid spaces [1] demonstrate LEAN 4's capacity for large-scale mathematical formalization. The Odd Order Theorem [5] and the Kepler Conjecture [6] are landmark formalizations in Coq and HOL Light, respectively.

Prime-producing formulas. The UGP prime formula (7) is a two-parameter family constrained by the Diophantine condition (6). This places it in the tradition of prime-producing polynomials (Euler's $n^2 + n + 41$, the Bunyakovsky conjecture) and prime-generating sieves, though the constraint to ridges $2^n - 16$ is novel. Exhaustive computation finds 57 terms up to 10^8.

Companion work. The UGP framework was introduced in [9]; the present formalization provides the machine-checked foundation. The self-reference theorems are imported from two companion LEAN 4 libraries:

- **nems-lean** [10]: Lawvere fixed-point theorem in abstract form, Kleene recursion theorem, and Rogers' fixed-point theorem, all proved from a minimal self-referential interface (SRI).
- **aps-rice-lean** [8]: Rice's theorem and halting undecidability for acceptable programming systems, proved from the s-m-n theorem and a diagonal argument.

The companion **primes-in-greedy-B3** [11] provides a related B_3 prime result: every prime appears in the greedy multiplicative B_3 sequence (OEIS A079852).

16 Conclusion

We have presented ugp-lean, a self-contained Lean 4 formalization of the Universal Generative Principle. The artifact defines the UGP framework from first principles—ridge sieve, prime-lock criterion, RSUC, GTE orbit, Quarter-Lock, Turing universality—with a strict zero-sorry, zero-custom-axiom policy on the core proof path.

The formalization contributes several results not present in prior work: the orbit-from-T derivation (theorem 6.1), the general ridge remainder lock (theorem 7.1), the even-step c-invariance theorem (theorem 6.2), the UGP prime class with ten machine-checked instances (theorem 8.3), the divisor count unboundedness theorem (theorem 11.3), the exact divisor-count formula $\tau(2^n - 16) = 5 \cdot \tau(2^{n-4} - 1)$ (theorem 11.4), the precise formal statement of the mirror-dual conjecture (theorem 11.5) with five machine-verified witness pairs, and the mirror shift algebra (section 12): the universal law $c_1(b_2, q_2) - c_1(q_2, b_2) = b_1(q_2 - b_2)$ and the shared residue $c_1 \equiv 20 \pmod{b_1}$ for all mirror-dual pairs at all ridge levels.

What remains open. The ten conjectures in table 2 are formally stated but unproved. The mirror-dual conjecture (theorem 11.5) and the UGP prime infinitude conjecture (theorem 8.5) are the most interesting: both are structurally analogous to the twin prime conjecture and would require new sieve-theoretic techniques. Dirichlet-series analysis of the ridge divisor function and Markov mixing properties of the GTE trajectory are natural directions for future formalization.

Reproducibility. The artifact is publicly available at `https://github.com/novaspivack/ugp-lean`. Building requires Lean 4 v4.29.0-rc6 and Mathlib v4.29.0-rc6. The command `lake build` compiles the entire library; no external tools beyond Lean 4 and its package manager are needed.

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Table 3: Complete theorem inventory. "Novel" indicates results new to this formalization; "General" indicates level-independent results. All theorems have zero `sorry` on the core proof path.

Theorem / Lean name	**Module**	**Paper ref**	**Novel?**
Core Classification (§5)			
`theoremA_general`	Classification.TheoremA	Thm A	—
`theoremB`	Classification.TheoremB	Thm B	—
`rsuc_theorem`	Classification.RSUC	RSUC	—
`mdl_selects_LeptonSeed`	Classification.TheoremB	MDL	—
`rsuc_formal`	Classification.FormalRSUC	RSUC	—
`rsuc_canon`	Classification.FormalRSUC	RSUC	—
`strengthening_cannot_add`	Classification.Monotonic	—	—
Ridge & Sieve (§5)			
`ridgeSurvivors_10`	Compute.Sieve	§5	—
`ridge_remainder_lock`	Core.RidgeRigidity	Lem m_2	—
`exclude_16` ... `exclude_63`	Compute.ExclusionFilters	§5	—
GTE Orbit (§6)			
`orbit_determined_by_T`	GTE.UpdateMap	Prop 5.1	**Yes**
`update_map_produces_orbit`	GTE.UpdateMap	Prop 5.1	**Yes**
`even_step_c_invariance`	GTE.UpdateMap	—	**Yes**
`c3_strict_eq_c2_at_n10`	GTE.UpdateMap	—	**Yes**
`gen1_mersenne_isolation`	GTE.UpdateMap	—	**Yes**
`mersenne_entanglement_gen`	GTE.UpdateMap	—	**Yes**
General-Level Theorems (§7)			
`ridge_remainder_lock_gen`	GTE.UpdateMap	Lem m_2	**General**
`quotient_gap_all`	GTE.GeneralTheorems	gap-13	**General**
`mirror_b1_invariance`	GTE.UpdateMap	—	**Yes**
`mersenne_gcd_identity`	GTE.UpdateMap	—	Mathlib
`alpha_echo`	GTE.GeneralTheorems	α-echo	**Checked**
`b_linear_growth`	GTE.GeneralTheorems	b-growth	—
`c_values_strictly_mono`	GTE.GeneralTheorems	c-mono	—
Quarter-Lock & Kernel (§9)			
`quarterLockLaw`	QuarterLock	§3	—
`quarterLockStability`	QuarterLock	§3	—
`k_L2_eq`	ElegantKernel	§3	—
`L_model_from_gauge_struct`	LModelDerivation	§3	—
Universality & Self-Reference (§10)			
`ugp_is_turing_universal`	Universality.TuringUniv	§4	—
`ugp_lawvere_fixed_point`	SelfRef.LawvereKleene	Lawvere	Import
`ugp_kleene_recursion_thm`	SelfRef.LawvereKleene	Kleene	Import
`ugp_rice_theorem`	SelfRef.RiceHalting	Rice	Import
`ugp_halting_undecidable`	SelfRef.RiceHalting	Halting	Import
UGP Primes (§8)			
`IsUGPPrime`	GTE.UGPPrimes	Def 8.1	**Yes**
`ugp_primes_exist`	GTE.UGPPrimes	—	**Yes**
`ten_ugp_primes`	GTE.UGPPrimes	—	**Yes**
`ugp_primes_at_three_lvls`	GTE.UGPPrimes	—	**Yes**
`c1Val_eq_c1FromPair`	GTE.UGPPrimes	—	**Yes**
`UGPPrimeInfinitudeConj`	GTE.UGPPrimes	—	**Yes**
Mirror-Dual Conjecture (§11)			
`card_divisors_ridge_unb`	GTE.MirrorDualConj	—	**Yes**
`tau_ridge_exact`	GTE.MirrorDualConj	Thm 11.4	**Yes**
`coprime_pow2_mersenne`	GTE.MirrorDualConj	—	**Yes**
`five_mirror_dual_pairs`	GTE.MirrorDualConj	—	**Yes**
`MirrorDualConjecture`	GTE.MirrorDualConj	—	**Yes**
Mirror Shift Algebra (§12)			

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